

User Manual for the ContourPlot Tool

1. Preface

1. 1. Software Overview

With the continued development of geological survey and engineering geology research, efficient processing and visualization of geological data have become increasingly important. Traditional geological data processing methods often suffer from low efficiency, complex operation, and poor data timeliness, making it difficult to meet the practical needs of modern geological research and engineering projects. This software was developed to provide users with an efficient tool for automatically generating contour profile plots of geological target parameters, such as velocity, temperature, and pressure. With this software, users can process, analyze, and visualize geological data more conveniently and accurately, thereby significantly improving work efficiency and supporting more timely decision-making and analysis.

1. 2. Target Users

This software is mainly intended for geological exploration engineers, geological researchers, engineering geologists, and other professionals in related fields, especially users who need to process and analyze large volumes of geological data. It is particularly suitable for data analysis and decision support in geological surveys, engineering exploration, and related fields.

1. 3. Main Functions

This software mainly provides automatic generation of contour plots.

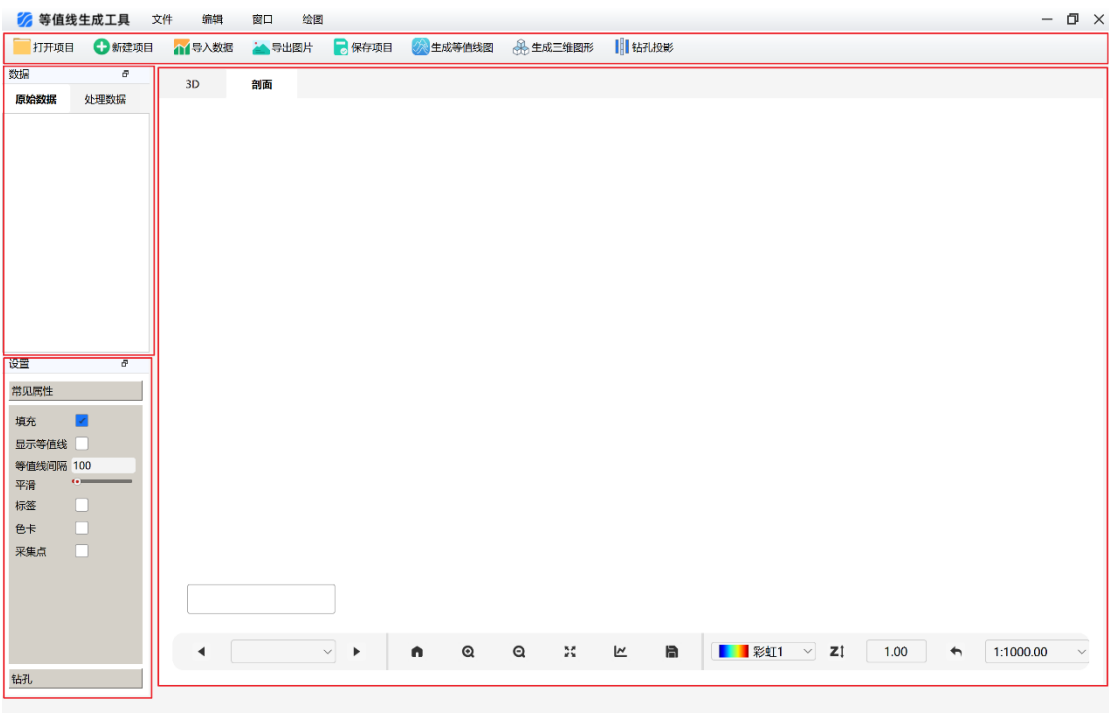
1. 4. How to Use This Manual

This user manual will help you quickly understand and use the functions of the software. It is designed to help you get started quickly and make efficient use of the available functions. Whether you are a new user or an experienced professional, this manual provides comprehensive operational guidance.

2. Basic Operations

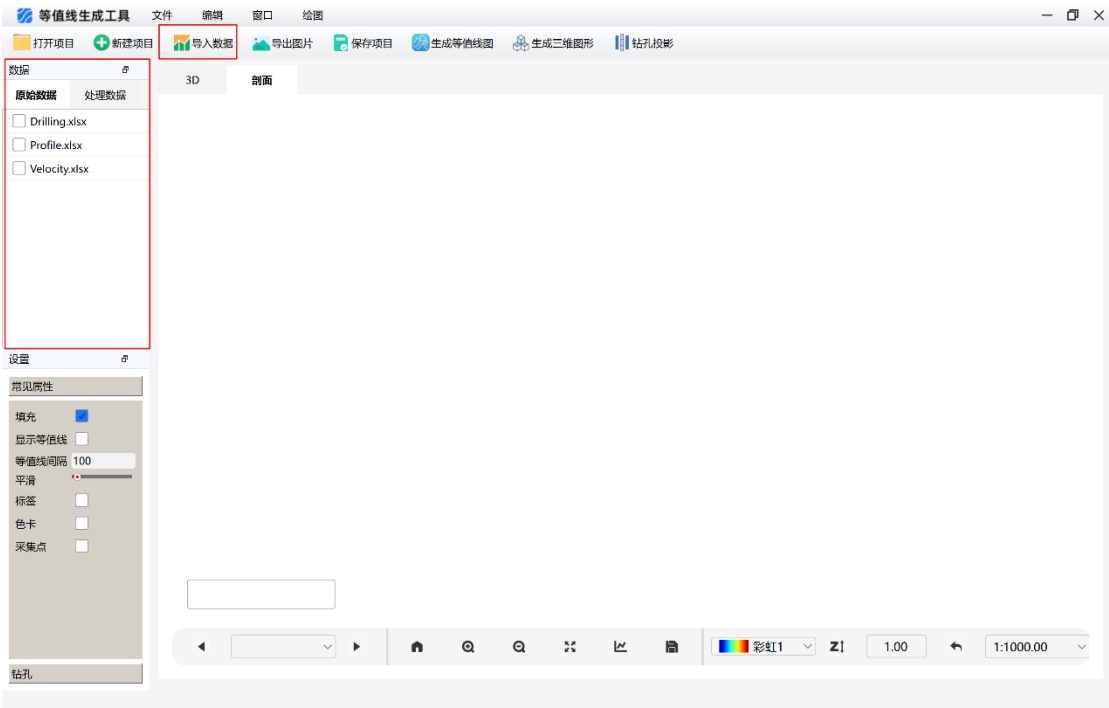
2. 1. Main Interface

The main interface of the software consists of four modules: the toolbar, data panel, settings panel, and image panel.



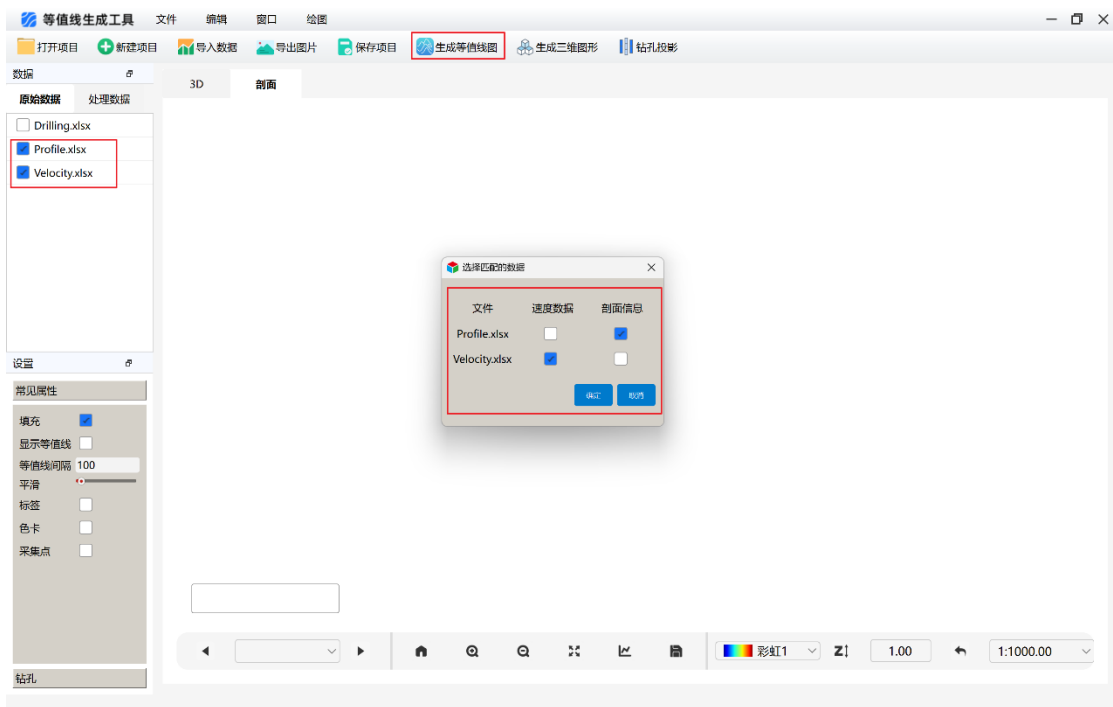
2. 2. File Management

Click the “Import Data” button on the toolbar to import data into the “Data Panel”. Double-click the data to view it.

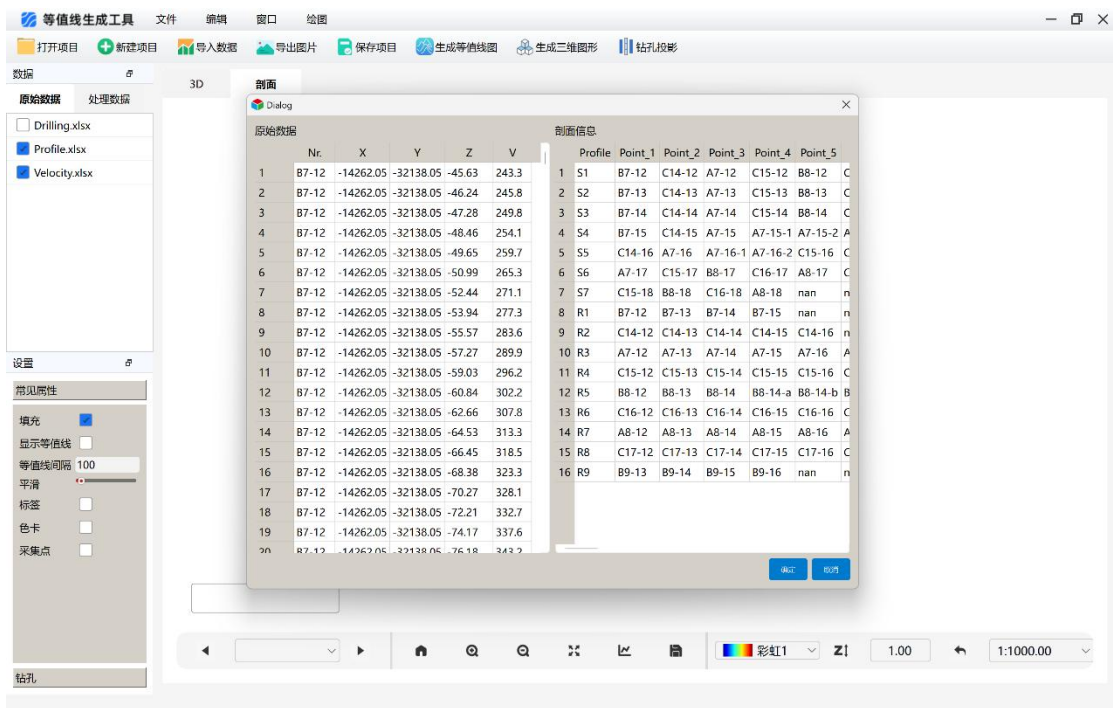


2. 3. Steps for Generating 2D and 3D Contour Plots

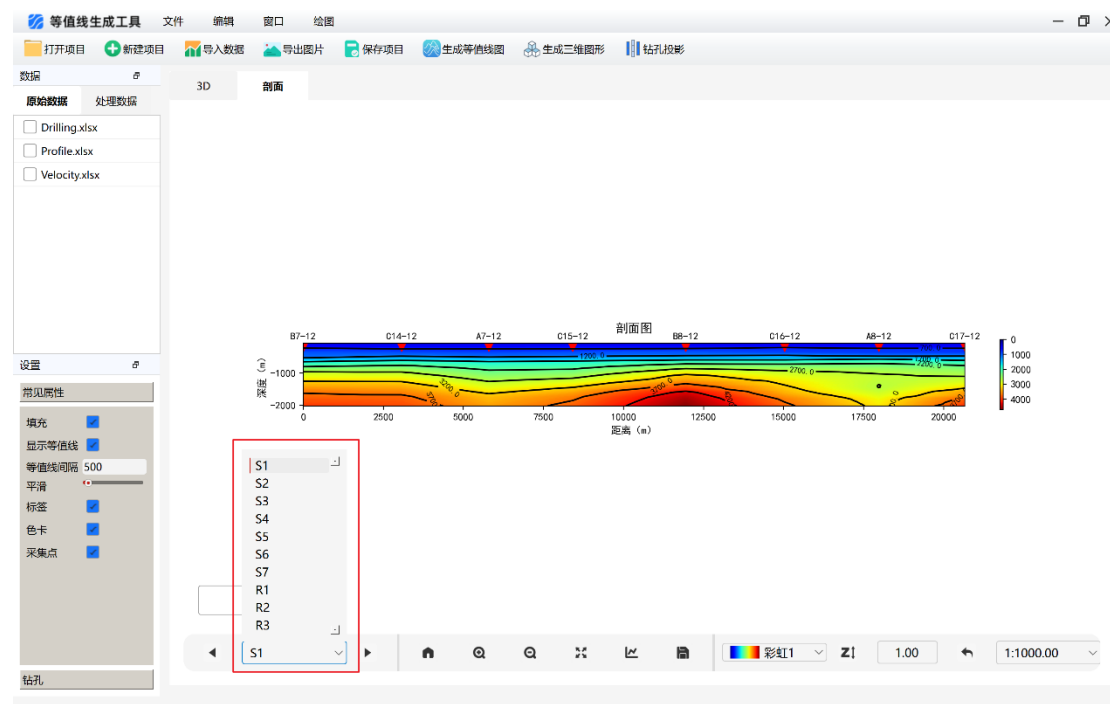
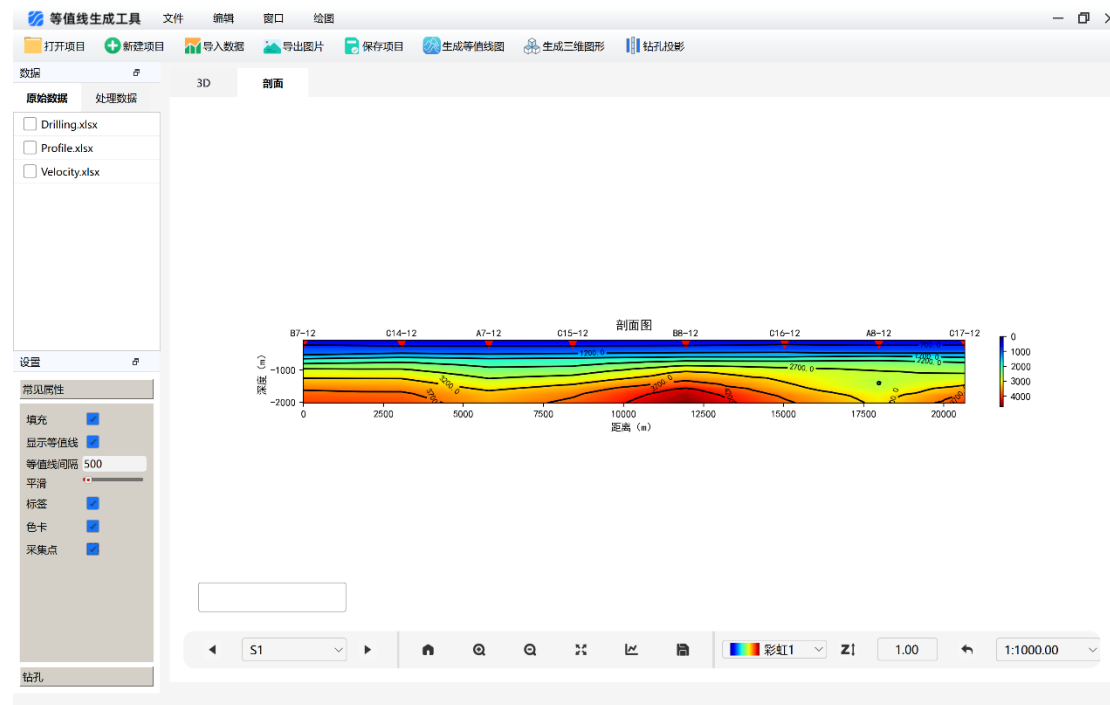
1. Select the relevant data, including “Raw Data” and section information. Click “Generate Contour Plot”, select the check boxes in the pop-up window to match the data, and then click “OK” .



2. Preview the data. After confirming that the data are correct, click “OK” . The relevant sections will then be generated automatically.

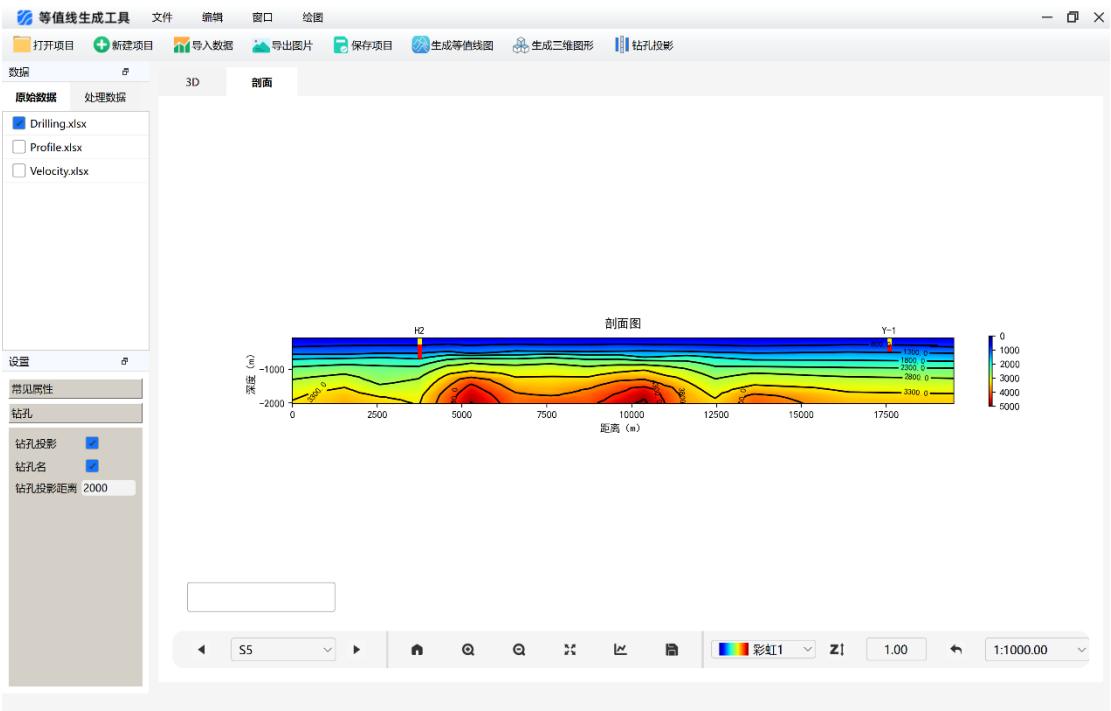


3. Modify section parameters: the “Settings Box” includes “General Properties” and “Boreholes”. Users can modify section parameters through the general properties. In the “Bottom Toolbar”, users can switch between sections using the left and right buttons or by selecting a section from the drop-down list.

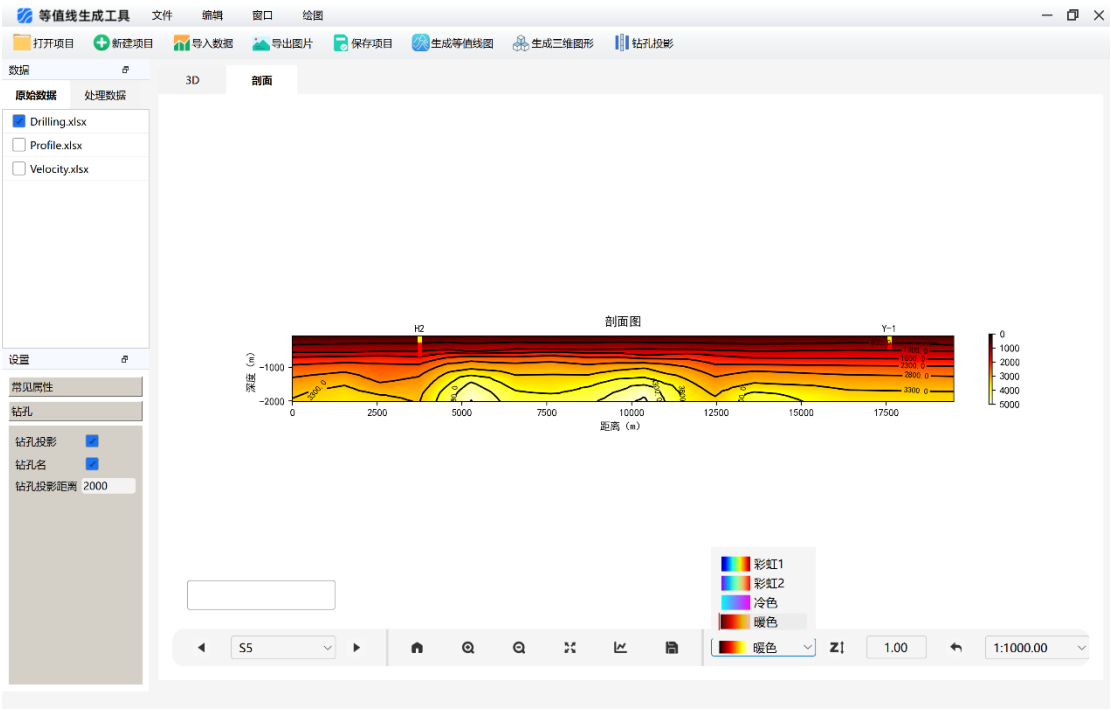


4. Add borehole projection: after importing borehole information, select the data and click “Borehole Projection”. Select “Borehole Name” to display borehole

names. The “Borehole Projection Distance” can be used to adjust the distance between the boreholes and the section.

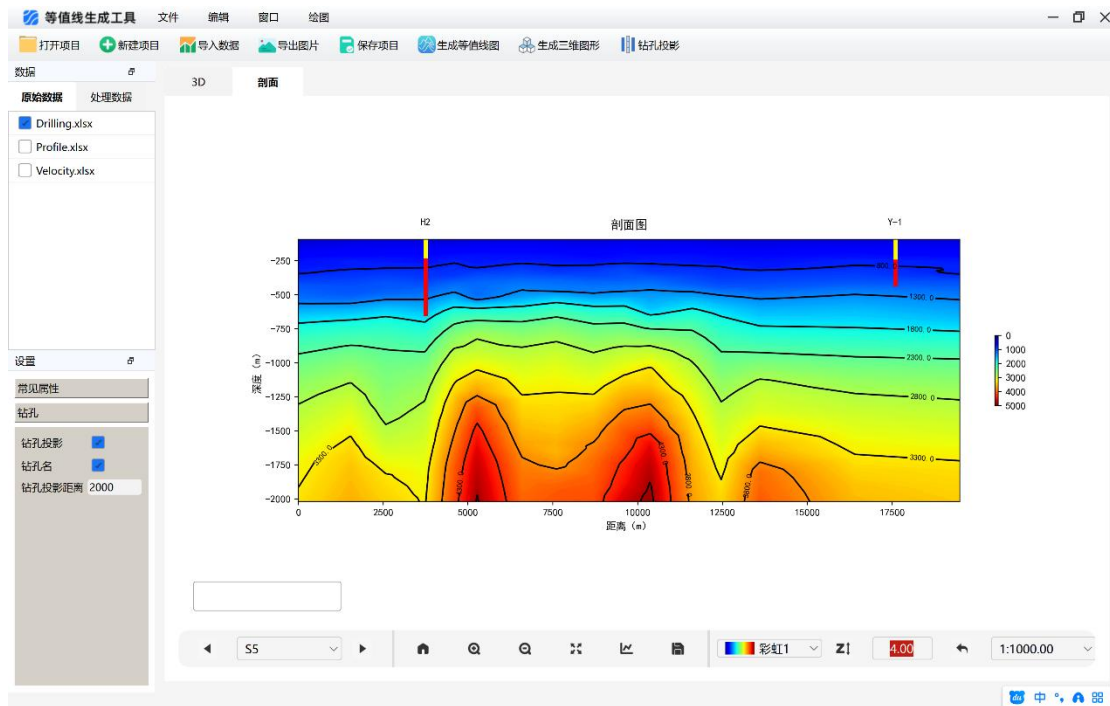


5. Adjust the color map: by default, the color map is set to rainbow. Users can modify and replace it as needed.

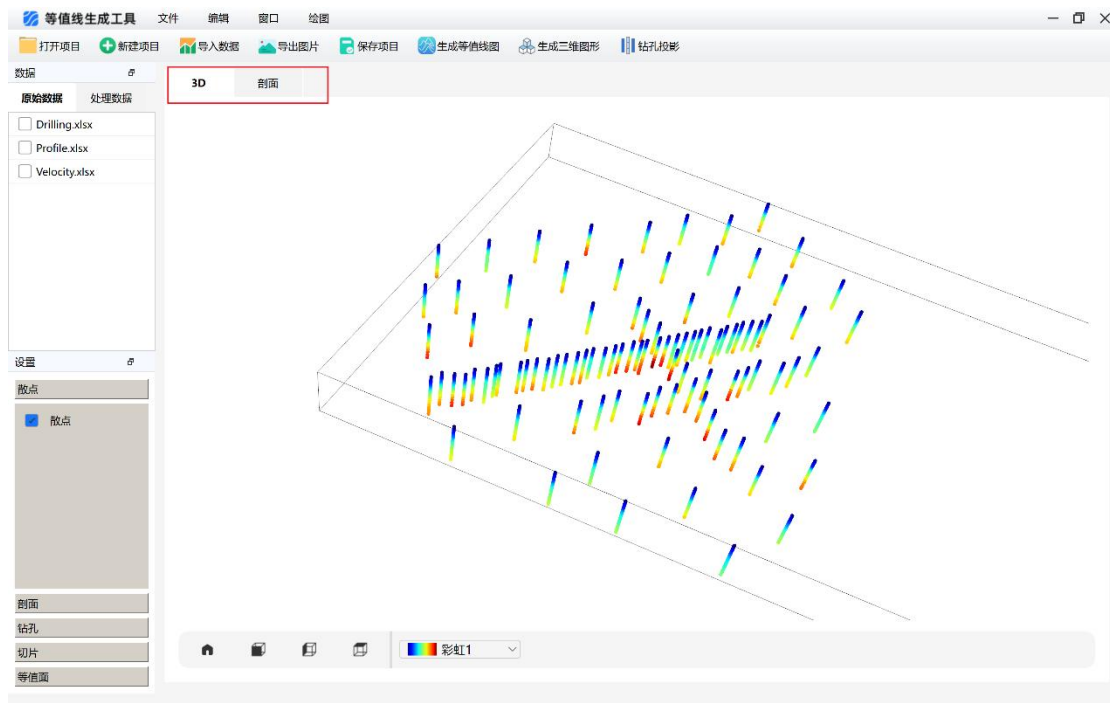


6. Adjust the aspect ratio: by default, the image aspect ratio is 1:1. Users can

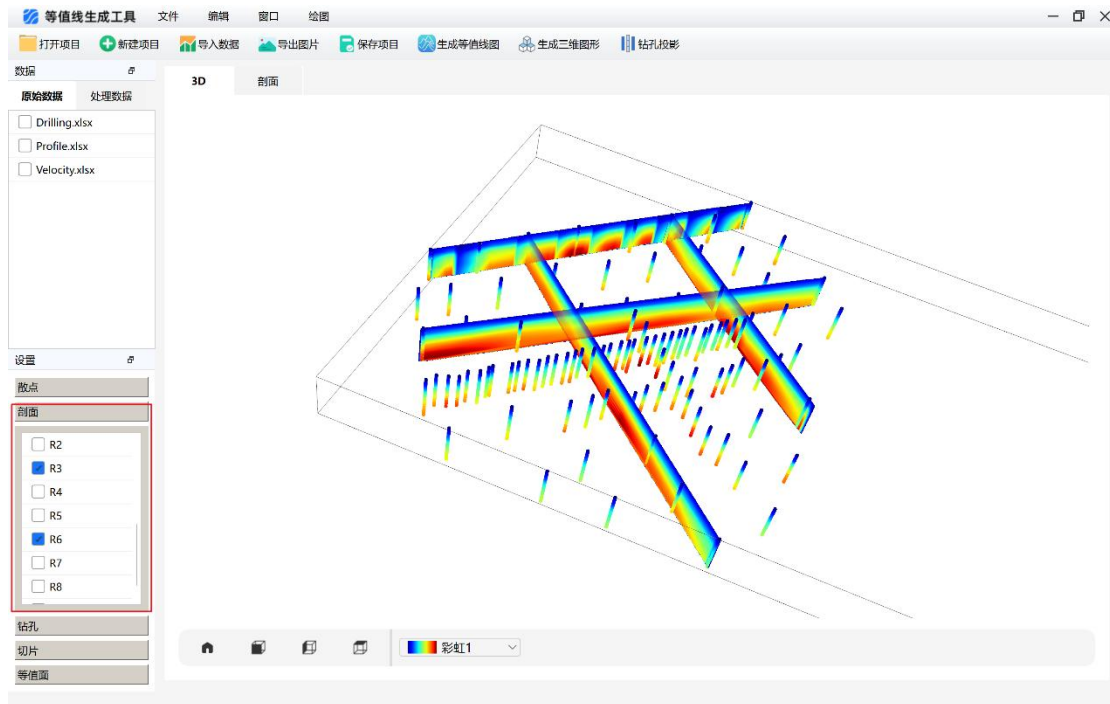
adjust the ratio by entering the desired value manually.



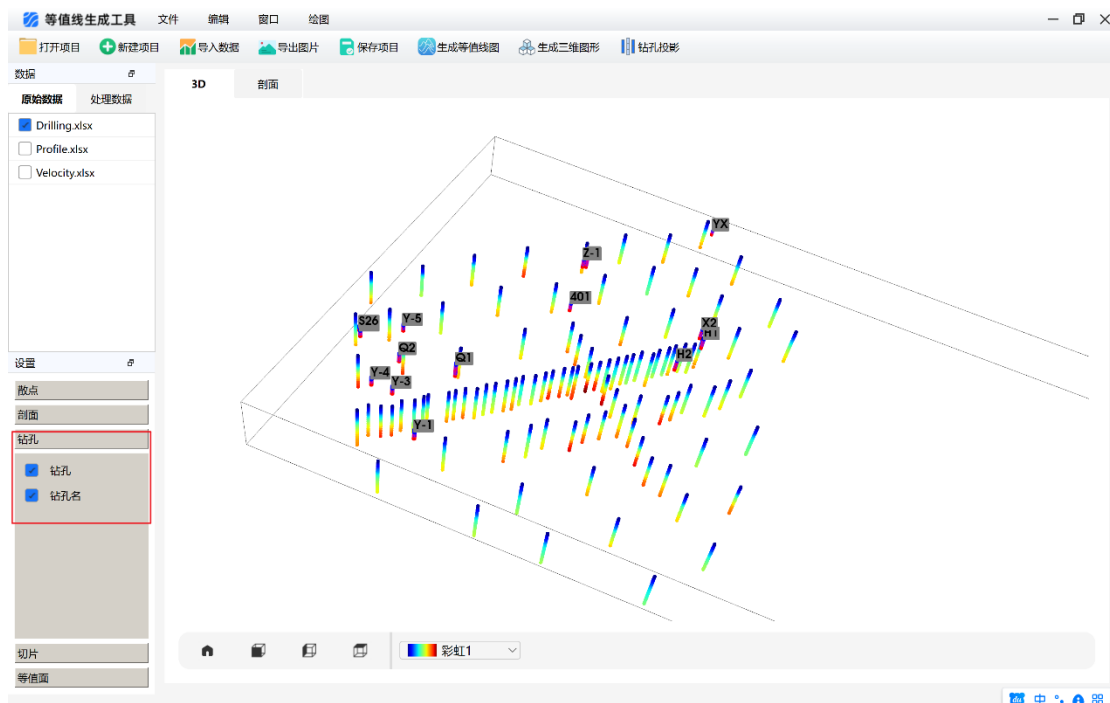
7. Click the “3D” button at the top of the “Image Panel” to switch to the 3D interface. The scatter plot is drawn by default.



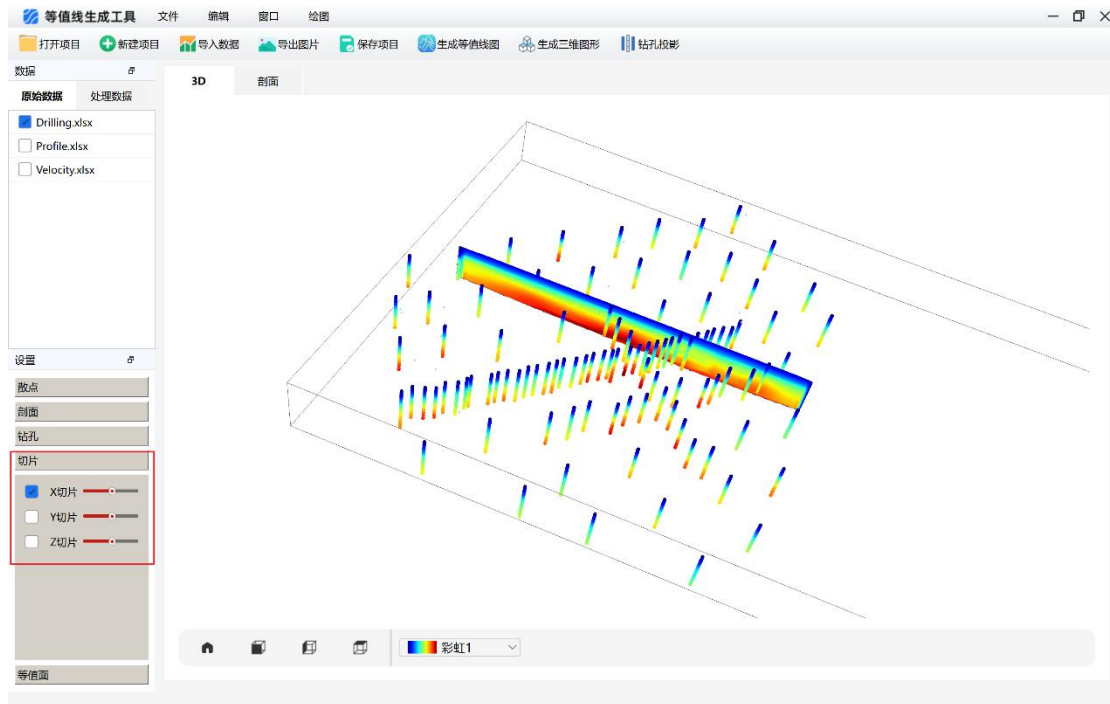
8. Generate 3D sections: on the section page in the “Settings Panel”, users can select the required sections to display them in 3D space.



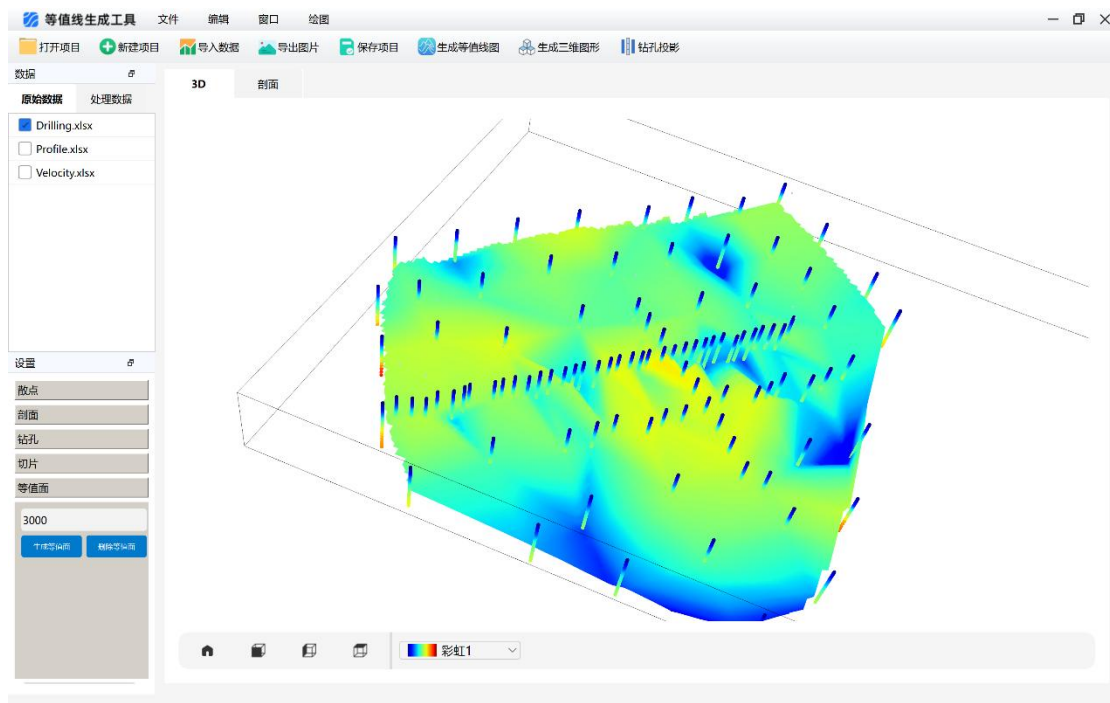
9. Display boreholes in 3D space: after selecting borehole data in the “Data Panel”, select “Borehole” in the “Settings Panel” to generate boreholes in 3D space.



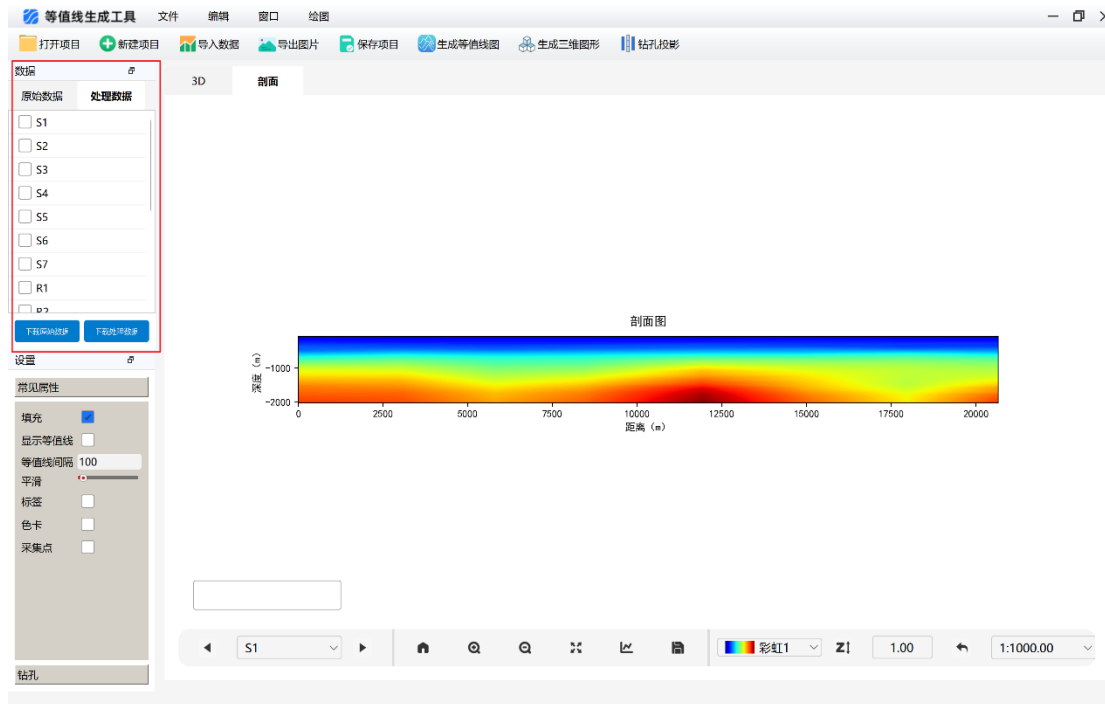
10. Generate slices: select “X Slice” in the “Settings Panel” to generate a slice in the X direction in 3D space. The slice position can be adjusted by dragging the slider.



11. Generate an isosurface: go to the “Isosurface” page in the “Settings Panel”, enter the isosurface value, and click “Generate Isosurface” or press Enter to generate the isosurface.



12. Download data: on the “Processed Data” page in the “Data Panel”, filtered data for each section are generated. Users can download the relevant data.



3. Data Examples

Raw data format: as shown in the figure below, the raw data format contains five columns: ID, X coordinate, Y coordinate, depth, and geological target parameter.

	编号	横坐标	纵坐标	深度	速度值
1	B7-12	-14262.05	-32138.05	-45.63	243.3
2	B7-12	-14262.05	-32138.05	-46.24	245.8
3	B7-12	-14262.05	-32138.05	-47.28	249.8
4	B7-12	-14262.05	-32138.05	-48.46	254.1
5	B7-12	-14262.05	-32138.05	-49.65	259.7
6	B7-12	-14262.05	-32138.05	-50.99	265.3
7	B7-12	-14262.05	-32138.05	-52.44	271.1
8	B7-12	-14262.05	-32138.05	-53.94	277.3
9	B7-12	-14262.05	-32138.05	-55.57	283.6
10	B7-12	-14262.05	-32138.05	-57.27	289.9
11	B7-12	-14262.05	-32138.05	-59.03	296.2
12	B7-12	-14262.05	-32138.05	-60.84	302.2
13	B7-12	-14262.05	-32138.05	-62.66	307.8
14	B7-12	-14262.05	-32138.05	-64.53	313.3
15	B7-12	-14262.05	-32138.05	-66.45	318.5
16	B7-12	-14262.05	-32138.05	-68.38	323.3
17	B7-12	-14262.05	-32138.05	-70.27	328.1
18	B7-12	-14262.05	-32138.05	-72.21	332.7
19	B7-12	-14262.05	-32138.05	-74.17	337.6
20	B7-12	-14262.05	-32138.05	-76.18	343.2
21	B7-12	-14262.05	-32138.05	-78.31	349.7
22	B7-12	-14262.05	-32138.05	-80.53	357.7

Section information: the section information format is shown in the figure below. It includes the section ID and the measurement points included in each section.

[illegible]